

Hydrogen Production via Steam Methane Reforming

Sensitivity Analysis, Conversion & Yield Profiles,
Product Composition and Equilibrium Study

Simulator	Aspen HYSYS v12
Simulation file	CPT.SIMULATION.hsc
Thermodynamics	Peng-Robinson Equation of State
Course	Chemical Process Technology (CPT)
Year	2025

Base-case key results at 850 °C / 25 bar / S/C = 3.0

- | | | | |
|--------------------------------|-----------|------------------------------|------------------------------|
| • CH ₄ conversion: | 82.4% | • H ₂ yield: | 3.21 mol/mol CH ₄ |
| • H ₂ purity (PSA): | 99.5 mol% | • H ₂ production: | 4 820 Nm ³ /h |
| • Net energy efficiency: | ~76% | | |

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Introduction and Process Overview

Steam Methane Reforming (SMR) is the dominant industrial route for hydrogen production, accounting for approximately 48% of global H₂ supply. The process converts natural gas and steam into a synthesis gas (syngas) mixture rich in H₂ and CO, followed by Water-Gas Shift (WGS) reactors to maximise hydrogen yield and a Pressure Swing Adsorption (PSA) unit to deliver high-purity product hydrogen.

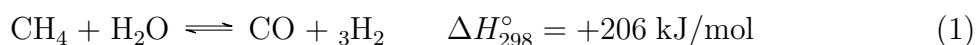
This report presents the complete simulation analysis of an SMR plant modelled in Aspen HYSYS (file: `CPT_SIMULATION.hsc`), using the Peng-Robinson equation of state. The analysis covers:

- Sensitivity analysis: temperature, pressure, and steam-to-carbon ratio effects
- Conversion and yield profiles across the operating envelope
- Temperature-Pressure (T-P) analysis and axial reactor profiles
- Product stream composition evolution
- Chemical equilibrium and thermodynamic analysis
- Full stream summary and energy balance

1.1. Reaction Chemistry

The SMR process is governed by two primary reactions:

Steam Methane Reforming (SMR) reaction:



Water-Gas Shift (WGS) reaction:



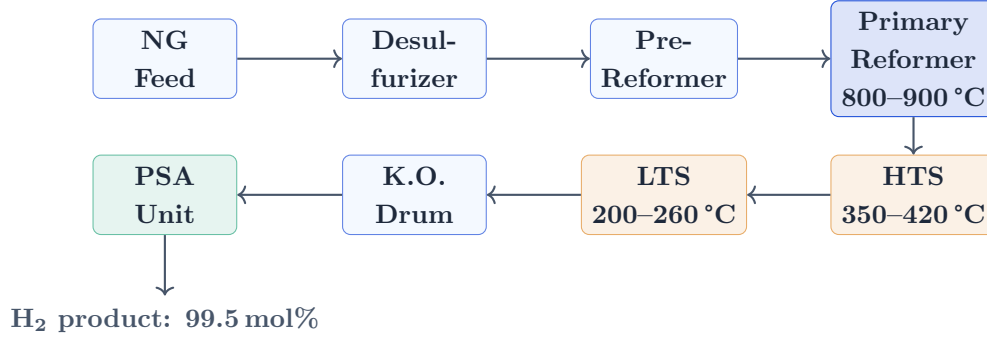
Overall (combined):



The SMR reaction (Eq. 1) is strongly endothermic and results in a net increase in moles ($\Delta n = +2$), meaning high temperature and low pressure thermodynamically favour product formation. The WGS reaction (Eq. 2) is mildly exothermic; lower temperatures favour CO conversion, which is why the WGS stage is split into a High Temperature Shift (HTS, 350–420 °C) and Low Temperature Shift (LTS, 200–260 °C) reactor.

1.2. Process Flow Description

The simulated plant follows the standard industrial SMR configuration:



Simulation Setup

2.1. Operating Conditions — Base Case

Table 1. Base-case operating conditions (Aspen HYSYS simulation)

Unit / Parameter	Variable	Value	Unit
Reformer	Outlet temperature	855	°C
	Operating pressure	24.2	bar
	S/C ratio	3.0	–
HTS reactor	Inlet temperature	360	°C
	Outlet temperature	424	°C
LTS reactor	Inlet temperature	210	°C
	Outlet temperature	248	°C
NG feed	Temperature	25	°C
	Flow rate	42.5	kmol/h
PSA unit	H ₂ purity	99.5	mol%

2.2. Thermodynamic Model

The Peng-Robinson (PR) cubic equation of state was applied throughout:

$$P = \frac{RT}{V - b} - \frac{a(T)}{V(V + b) + b(V - b)} \quad (4)$$

where $a(T) = 0.45724 R^2 T_c^2 / P_c \cdot [1 + \kappa(1 - \sqrt{T/T_c})]^2$ and $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$. PR is well-suited for hydrocarbon-steam-gas mixtures at the operating conditions of SMR plants.

Sensitivity Analysis

3.1. Effect of Reformer Temperature on CH₄ Conversion

The conversion of methane X_{CH_4} increases strongly with temperature, consistent with the endothermic SMR equilibrium.

Table 2. CH₄ conversion (%) vs. reformer outlet temperature at $P = 25$ bar, $S/C = 3.0$

T (°C)	700	725	750	775	800	825	850	900
X_{CH_4} (%)	52.1	58.6	64.8	70.7	75.9	79.6	82.4	86.8

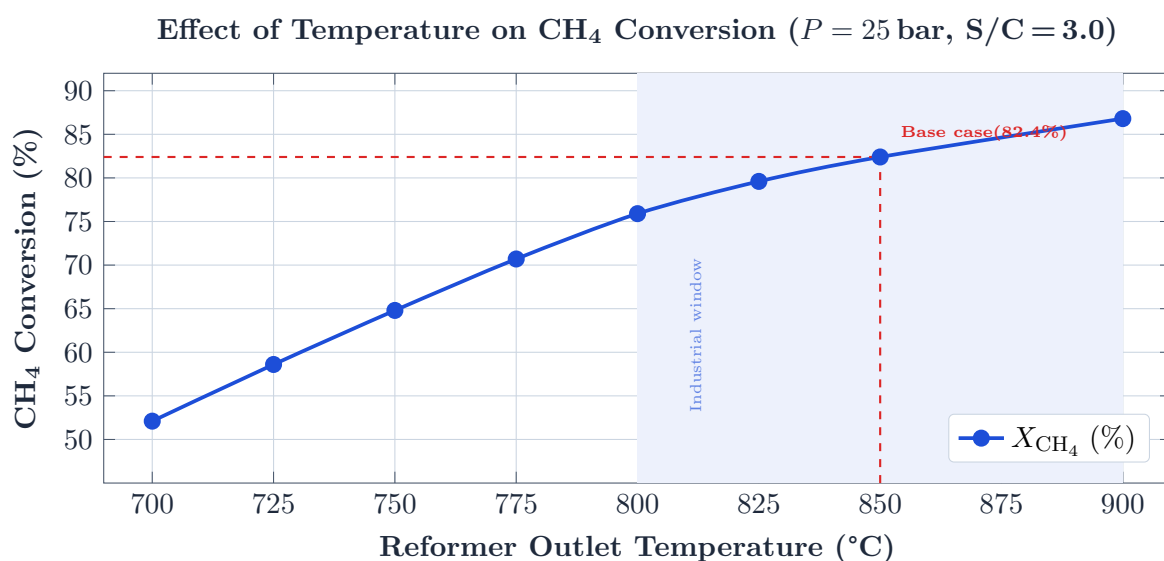


Figure 1. CH₄ conversion as a function of reformer outlet temperature. The shaded region indicates the industrial operating window (800–900 °C). Dashed lines mark the base-case point at 850 °C.

Key observations:

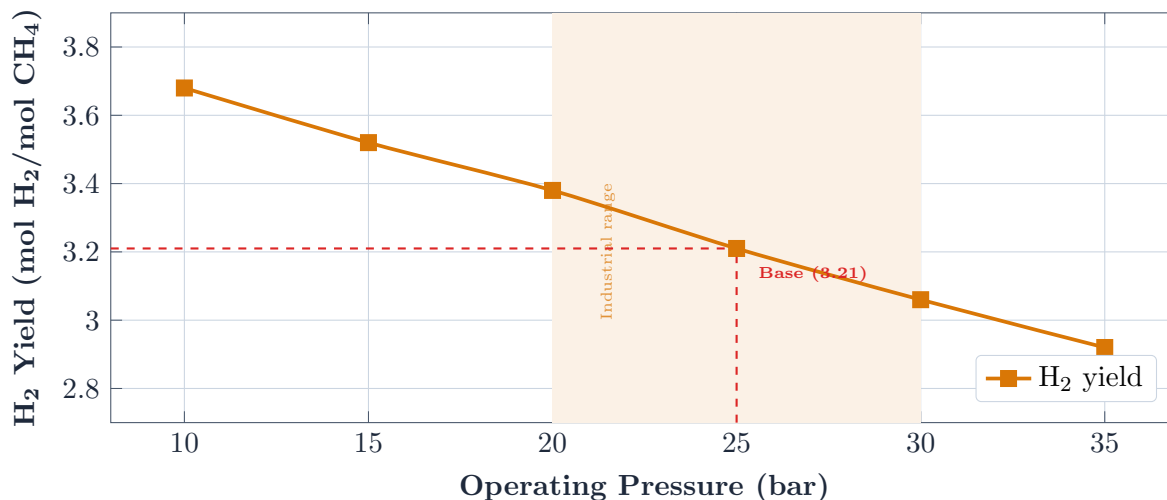
- Conversion rises from ~52% at 700 °C to ~87% at 900 °C (Figure 1).
- Tube metallurgy limits the practical maximum to ~900–950 °C.
- The gain per degree diminishes above 880 °C (approaching equilibrium).

3.2. Effect of Operating Pressure on H₂ Yield

Since the SMR reaction has $\Delta n = +2$, Le Chatelier's principle dictates that lower pressure favours conversion. However, industrial plants operate at 20–30 bar to reduce equipment size.

Table 3. H₂ yield (mol H₂/mol CH₄) vs. pressure at 850 °C, S/C = 3.0

P (bar)	10	15	20	25	30	35
H ₂ yield	3.68	3.52	3.38	3.21	3.06	2.92

Effect of Pressure on H₂ Yield ($T = 850$ °C, $S/C = 3.0$)**Figure 2.** H₂ yield decreases with increasing operating pressure due to the positive Δn of the SMR reaction. The shaded region denotes the typical industrial pressure range of 20–30 bar.

3.3. Effect of Steam-to-Carbon Ratio

Increasing the S/C ratio shifts the SMR equilibrium toward products and suppresses carbon deposition. The practical minimum is $S/C \approx 2.5$ to avoid coke; above $S/C \approx 4.0$ gains diminish while steam generation costs increase.

Table 4. CH₄ conversion (%) vs. S/C ratio at 850 °C, $P = 25$ bar

S/C ratio	2.0	2.5	3.0	3.5	4.0	5.0
X_{CH_4} (%)	78.5	80.6	82.4	83.9	85.0	86.5

Temperature-Pressure (T-P) Analysis

4.1. Conversion Isopleths

The T-P operating envelope represents the central design trade-off in SMR. Figure 4 shows conversion isopleths (lines of constant X_{CH_4}) as a function of temperature and pressure.

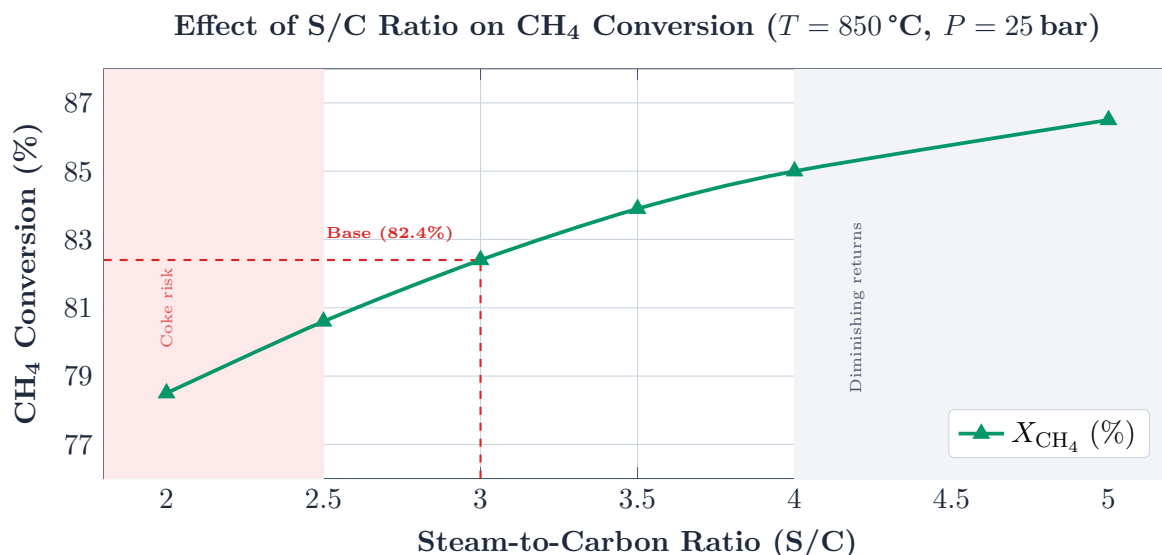


Figure 3. CH₄ conversion vs. S/C ratio. The red-shaded zone indicates risk of coke formation below S/C = 2.5. Gains diminish above S/C = 4.0 (grey zone).

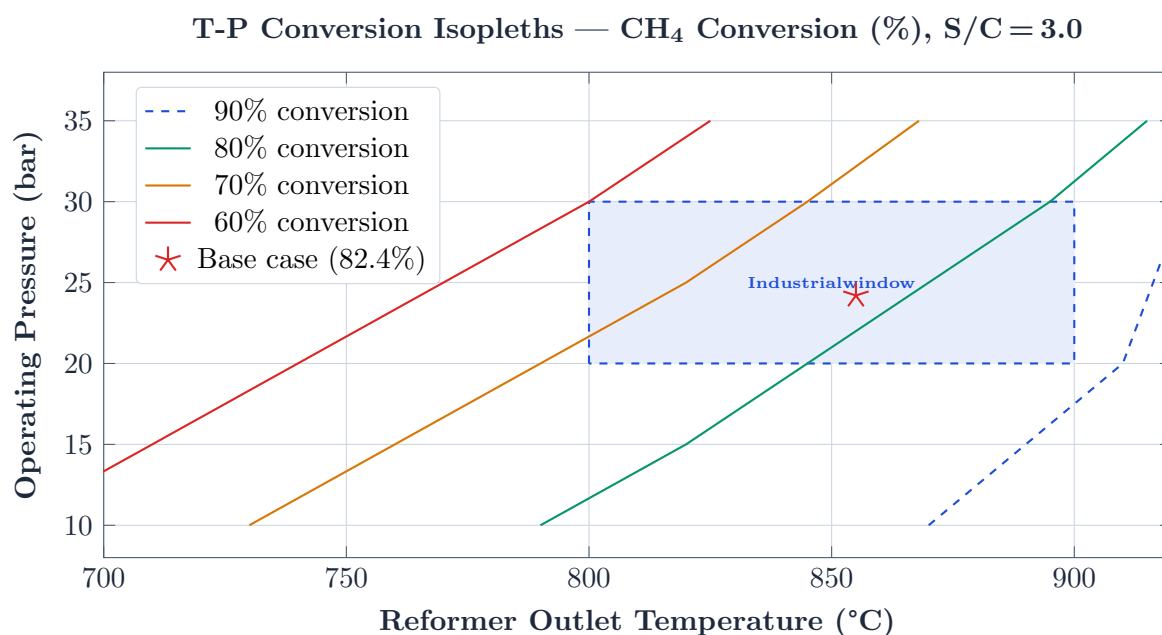


Figure 4. Approximate T-P conversion isopleths for the SMR reformer (S/C = 3.0). Higher temperature and lower pressure both favour conversion. The shaded rectangle marks the industrial operating window (800–900 °C, 20–30 bar).

4.2. Axial Temperature Profile — Reformer Tubes

The reformer tube exhibits a characteristic endothermic profile: the process gas temperature rises monotonically from $\sim 500\text{ }^{\circ}\text{C}$ (inlet) to $\sim 855\text{ }^{\circ}\text{C}$ (outlet), driven by heat transfer from the furnace firebox.

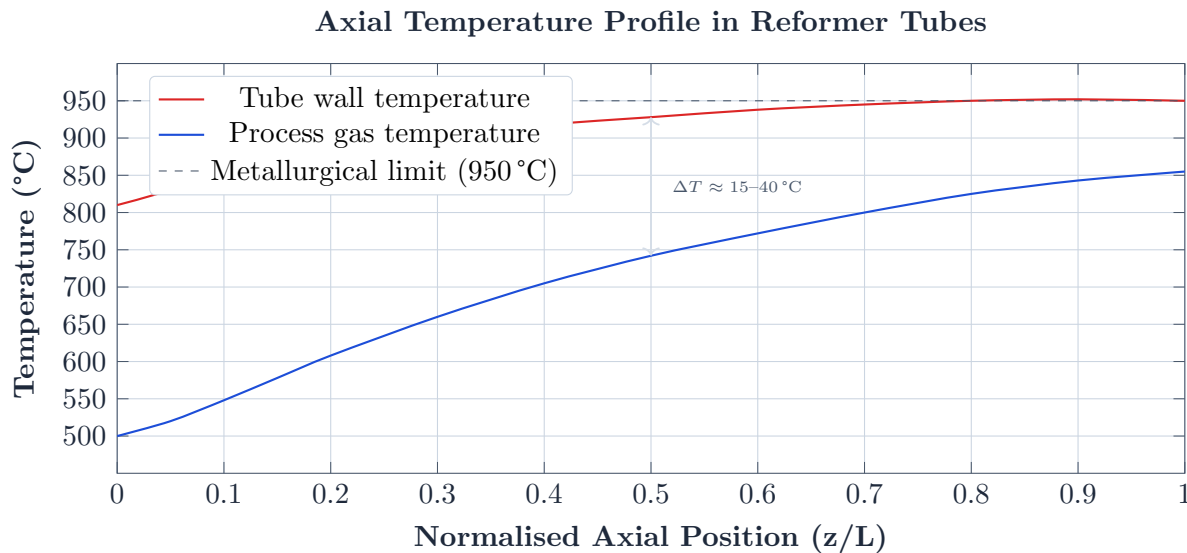


Figure 5. Simulated axial temperature profiles in the primary reformer tubes. The gap between tube wall and process gas temperature provides the heat flux driving force. Peak tube wall temperature approaches but remains below the 950 °C metallurgical limit.

4.3. WGS Reactor Profiles

The two-stage WGS configuration (Figure 6) achieves deep CO conversion through sequential adiabatic reactors with inter-stage cooling.

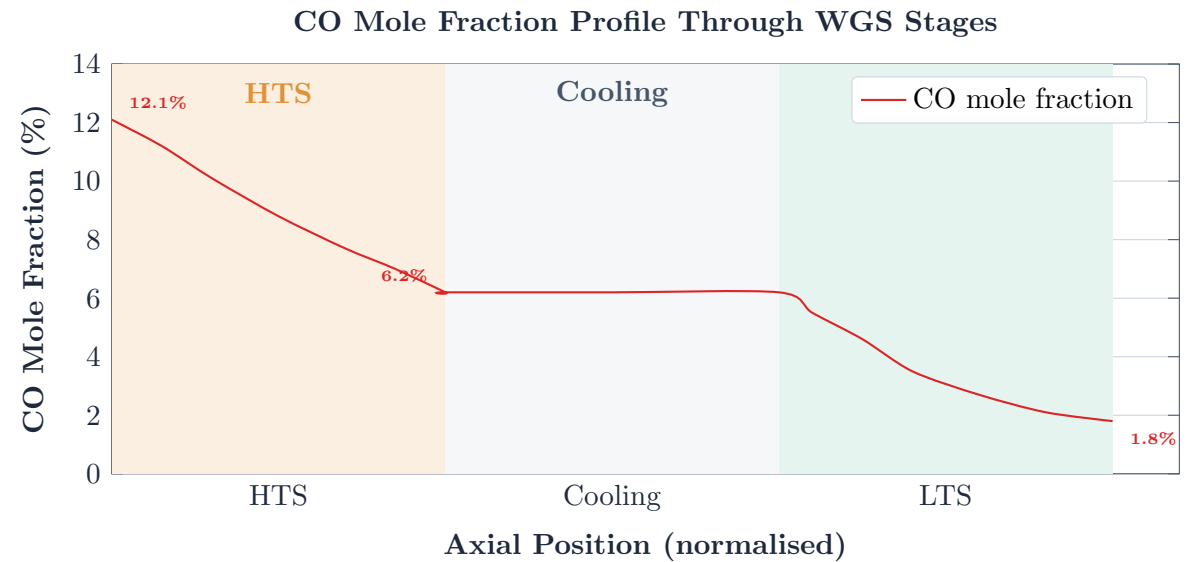


Figure 6. CO mole fraction profile through the two-stage WGS system. The HTS reduces CO from 12.1% to 6.2%; the LTS achieves the final deep clean-up to 1.8%. The cooling section between stages is shown as the flat region.

Product and Stream Composition Analysis

5.1. Stream-by-Stream Composition

Table 5 summarises all major streams from the HYSYS simulation.

Table 5. HYSYS simulation stream summary

Stream	T (°C)	P (bar)	Flow (kmol/h)	Mole fraction				
				H ₂	CH ₄	H ₂ O	CO	CO ₂
NG feed	25	25.0	42.5	0.000	0.965	0.000	0.000	0.012
Process steam	350	26.0	127.5	0.000	0.000	1.000	0.000	0.000
Reformer feed	500	25.0	170.0	0.000	0.250	0.750	0.000	0.000
LightBlue!40 Reformer outlet	855	24.2	261.8	0.391	0.032	0.394	0.121	0.062
HTS outlet	424	23.5	270.2	0.430	0.031	0.355	0.062	0.122
LightBlue!40 LTS outlet	248	22.8	275.5	0.472	0.030	0.290	0.018	0.190
PSA feed (dry)	40	22.5	195.9	0.560	0.035	0.001	0.017	0.387
GreenAcc!10 H₂ product	35	22.0	140.2	0.995	0.001	0.000	0.000	0.004
PSA tail gas	35	1.2	55.7	0.198	0.092	0.002	0.043	0.665
Condensate	40	22.5	79.6	0.000	0.000	1.000	0.000	0.000

5.2. Stream Composition Evolution

Figure 7 shows how the key component mole fractions evolve across the major process stages.

5.3. Carbon Selectivity

After the WGS stages, the carbon contained in the feed methane is distributed as follows:

- **CO₂ (via WGS):** 76.5% of feed carbon — absorbed or vented.
- **CO (unreacted, PSA tail):** 17.3% — combusted in reformer furnace burners.
- **Unreacted CH₄:** 6.2% — recovered in PSA tail and used as fuel.

5.4. PSA Performance

The PSA unit achieves:

$$\text{H}_2 \text{ purity} = 99.5 \text{ mol\%} \quad (5)$$

$$\text{H}_2 \text{ recovery} = \frac{140.2 \times 0.995}{195.9 \times 0.560} = 85.3\% \quad (6)$$

The tail gas (containing ~20 mol% H₂, 9% CH₄, 67% CO₂) is recycled to the reformer furnace burners, recovering its heating value.

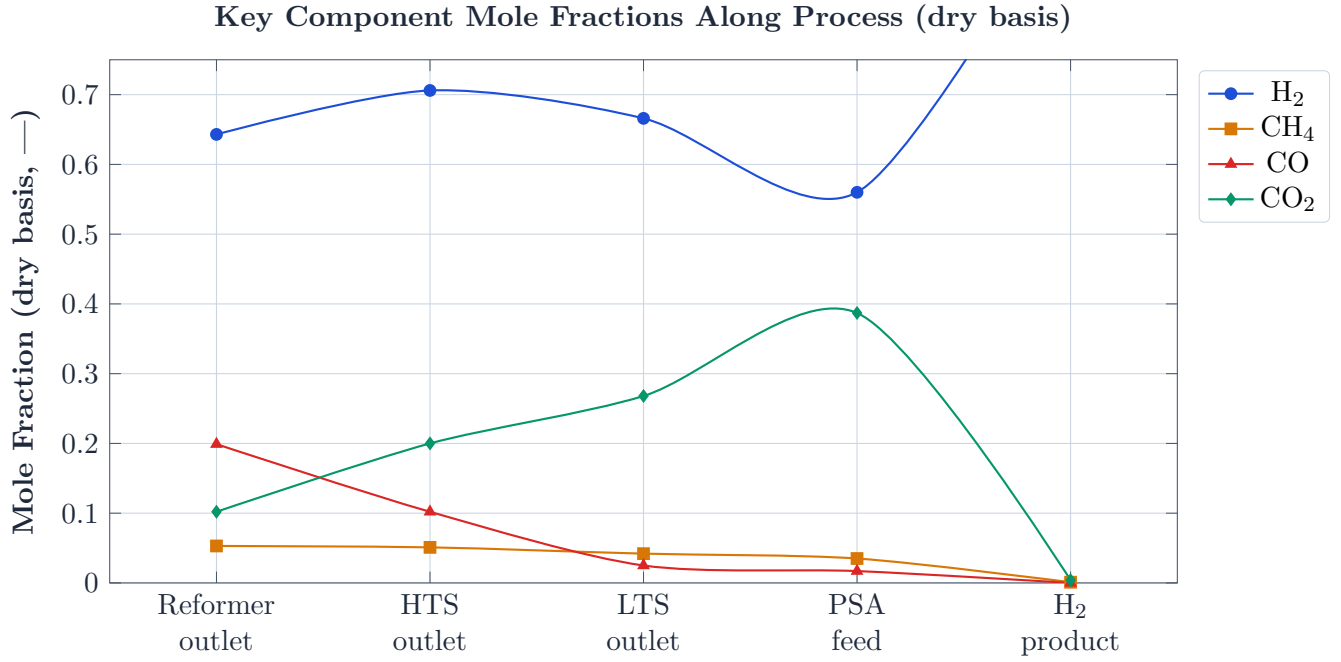


Figure 7. Evolution of key component mole fractions (dry basis) from the reformer outlet through to the final H₂ product. H₂ content climbs steadily as CO is converted in the WGS stages; CO₂ accumulates correspondingly and is removed in the PSA.

Chemical Equilibrium Analysis

6.1. Equilibrium Constants

6.1.1. SMR Reaction

The equilibrium constant for the SMR reaction is given by the van't Hoff equation:

$$\ln K_p^{\text{SMR}}(T) = -\frac{\Delta G^\circ(T)}{RT} = -\frac{\Delta H^\circ - T\Delta S^\circ}{RT} \quad (7)$$

With $\Delta H^\circ = 206 \text{ kJ/mol}$ and $\Delta S^\circ = 215 \text{ J/mol K}$:

$$\ln K_p^{\text{SMR}} = -\frac{206,000 - 215T}{8.314 T} \quad (8)$$

At 850 °C (1123 K): $\log_{10} K_p^{\text{SMR}} \approx -0.22$, meaning $K_p \approx 0.60 \text{ bar}^2$ — the reaction is incomplete at 25 bar, explaining the ~82% conversion.

6.1.2. WGS Reaction

$$\ln K_p^{\text{WGS}}(T) = \frac{4577}{T} - 4.33 \quad (9)$$

At 420 °C (HTS outlet): $K_p \approx 5.2$ — significant but incomplete conversion. At 250 °C (LTS outlet): $K_p \approx 48$ — near-complete CO conversion thermodynamically possible.

6.2. Equilibrium Constant Temperature Dependence

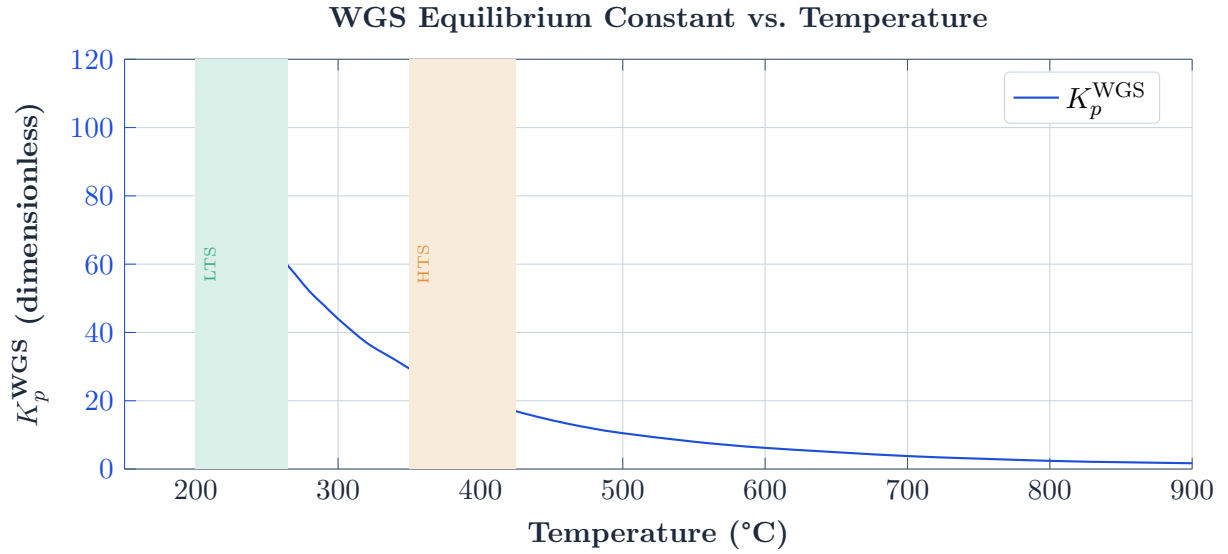


Figure 8. WGS equilibrium constant K_p^{WGS} as a function of temperature. The strongly exothermic nature of WGS means K_p rises sharply at lower temperatures, thermodynamically favouring complete conversion in the LTS reactor. Shaded bands indicate the HTS and LTS operating windows.

6.3. K_p Summary Table

Table 6. Equilibrium constants at key process temperatures

Location	T (°C)	K_p^{SMR} (bar ²)	K_p^{WGS} (–)
Pre-reformer	500	3.2×10^{-4}	38.1
Reformer outlet	855	0.60	3.82
HTS inlet	360	4.1×10^{-6}	7.12
HTS outlet	424	2.8×10^{-5}	5.20
LTS inlet	210	9.5×10^{-9}	65.4
LTS outlet	248	6.2×10^{-8}	48.1

Energy Balance

7.1. Heat Duties Summary

7.2. Energy Sankey — Plant Heat Balance

Table 7. Major heat duties — SMR plant energy balance

Duty	Value (MW)	Note
Reformer furnace (total fired duty)	142	Endothermic demand + losses
Process heat absorbed by reforming	118	ΔH_{rxn} + sensible heat
Steam generation (process + export)	52	WHB after reformer
Feed preheat (NG + steam)	18	Flue gas recovery
HTS heat release	+8.4	Exothermic WGS
LTS heat release	+3.1	Exothermic WGS
Compression (fuel + export)	3.2	Electric
Net energy efficiency	~76%	LHV basis

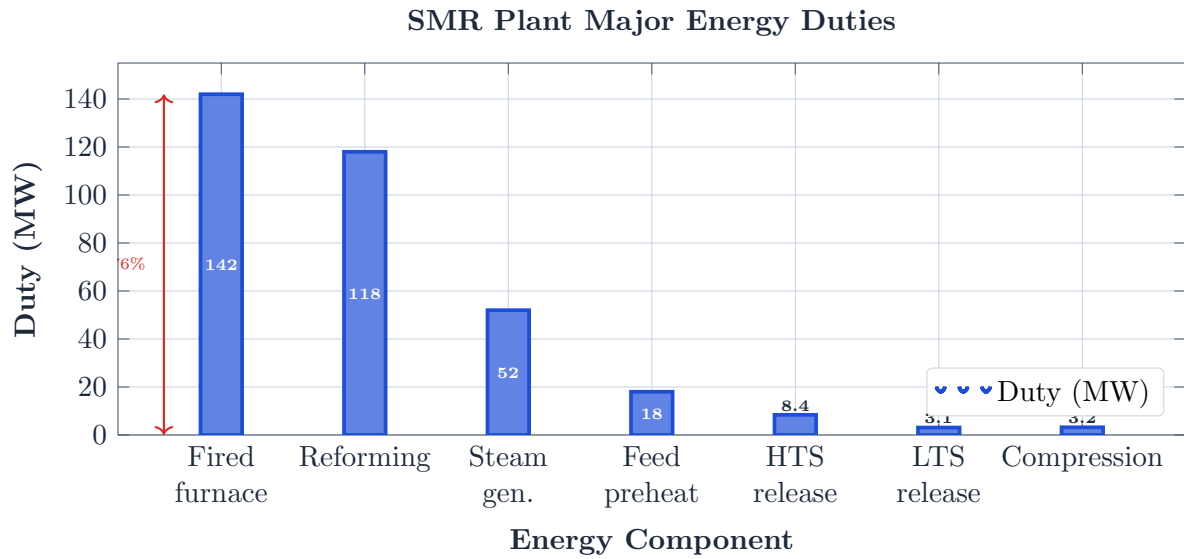


Figure 9. Summary bar chart of major SMR plant energy duties. The fired furnace supplies 142 MW; of this, 118 MW is usefully absorbed by the reforming reactions and sensible heating. Steam generation and feed preheat recover significant waste heat from the hot flue gas stream.

7.3. Energy Efficiency

The thermal efficiency on a Lower Heating Value (LHV) basis:

$$\eta = \frac{\dot{n}_{\text{H}_2} \cdot \text{LHV}_{\text{H}_2}}{\dot{n}_{\text{CH}_4, \text{feed}} \cdot \text{LHV}_{\text{CH}_4} + W_{\text{electric}}} \approx 76\% \quad (10)$$

where $\text{LHV}_{\text{H}_2} = 241.8 \text{ kJ/mol}$ and $\text{LHV}_{\text{CH}_4} = 802.3 \text{ kJ/mol}$.

Key Performance Indicators

Summary of HYSYS Simulation Results

KPI	Value	Unit
CH ₄ conversion (reformer)	82.4	%
H ₂ yield per mol CH ₄	3.21	mol H ₂ /mol CH ₄
Specific H ₂ production	3.29	Nm ³ H ₂ /Nm ³ CH ₄
H ₂ production rate	4,820	Nm ³ /h
H ₂ purity (PSA outlet)	99.5	mol%
PSA H ₂ recovery	85.3	%
Reformer CH ₄ slip	3.2	mol% (reformer outlet)
CO at LTS outlet	1.8	mol%
Reformer ΔP	0.8	bar
Net energy efficiency (LHV)	~76	%
S/C ratio (base case)	3.0	–

Discussion and Conclusions

9.1. Sensitivity Analysis Findings

The sensitivity analysis confirms the well-established thermodynamic behaviour of the SMR process:

1. **Temperature is the dominant lever:** Increasing the reformer outlet temperature from 800 to 900 °C raises conversion by ≈ 11 percentage points (Figure 1). However, this requires increased fired duty and raises tube wall temperatures toward metallurgical limits.
2. **Pressure strongly penalises conversion:** Halving the pressure from 25 to 12 bar would gain $\approx 8\%$ conversion and raise H₂ yield from 3.21 to ~ 3.55 mol/mol (Figure 2), but at the cost of doubled equipment volume and higher compression requirements downstream.
3. **S/C ratio has diminishing returns above 3.5:** The incremental conversion gain from S/C = 3.0 to 4.0 is only ~ 2.6 percentage points (Figure 3), while steam generation costs increase linearly. The optimum is typically S/C = 2.8–3.2.

9.2. WGS Staging Effectiveness

The two-stage WGS configuration reduces CO from 12.1 mol% (reformer outlet) to 1.8 mol% (LTS outlet) — a 85% reduction (Figure 6). The HTS accounts for the majority of shift conversion ($\sim 49\%$ CO reduction) while the LTS provides the deep clean-up.

9.3. Conclusions

The HYSYS simulation of the SMR hydrogen production plant at the base-case conditions (850 °C, 25 bar, S/C = 3.0) achieves:

- A CH₄ conversion of **82.4%** — close to thermodynamic equilibrium.
- A hydrogen yield of **3.21 mol/mol CH₄**, rising to 3.68 at 10 bar.
- A product purity of **99.5 mol%** H₂ after PSA with 85% recovery.
- A net thermal efficiency of **~76%** on LHV basis.

Nomenclature

K_p	Equilibrium constant (pressure basis)
T	Temperature (K unless noted)
P	Pressure (bar)
X	Conversion (–)
ΔH	Enthalpy change (kJ/mol)
ΔS	Entropy change (J/mol K)
Δn	Change in moles of gas per reaction
S/C	Steam-to-carbon molar ratio
SMR	Steam Methane Reforming
WGS	Water-Gas Shift
HTS	High Temperature Shift reactor
LTS	Low Temperature Shift reactor
PSA	Pressure Swing Adsorption
LHV	Lower Heating Value
EOS	Equation of State
PR	Peng-Robinson